

CIRCLE EQUATIONS

Digital SAT Style Practice Questions
DOCUMENT 1 — QUESTIONS ONLY

Easy 25% (10) | Medium 45% (18) | Hard 30% (12) | Total Questions: 40

Q1. • EASY

What is the center of the circle in the xy -plane defined by the equation $(x - 6)^2 + (y + 5)^2 = 64$?

- A) (6, -5)
- B) (6, 5)
- C) (-6, 5)
- D) (-6, -5)

Q2. • EASY

What is the radius of the circle in the xy -plane defined by the equation $(x + 8)^2 + (y - 3)^2 = 121$?

- A) 121
- B) 11
- C) 60.5
- D) 22

Q3. • EASY

A circle in the xy -plane has center $(-3, 8)$ and radius 5. Which of the following is an equation of this circle?

- A) $(x + 3)^2 + (y - 8)^2 = 5$
- B) $(x - 3)^2 + (y + 8)^2 = 25$
- C) $(x + 3)^2 + (y - 8)^2 = 25$
- D) $(x + 3)^2 + (y - 8)^2 = 10$

Q4. • EASY

What is the center of the circle in the xy -plane defined by the equation $x^2 + (y + 4)^2 = 36$?

- A) (0, 4)
- B) (4, 0)
- C) (-4, 0)
- D) (0, -4)

Q5. • EASY GRID-IN

A circle in the xy -plane has center $(2, 3)$ and passes through the point $(2, 9)$. What is the radius of the circle?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q6. • EASY

A circle in the xy -plane is defined by the equation $(x - 5)^2 + (y + 1)^2 = 81$. What is the diameter of this circle?

- A) 18
 - B) 9
 - C) 81
 - D) 40.5
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Q7. • EASY

A circle in the xy -plane is defined by the equation $(x + 2)^2 + (y - 6)^2 = 49$. What is the circumference of this circle?

- A) 7π
 - B) 14π
 - C) 49π
 - D) 28π
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Q8. • EASY

What is the center of the circle in the xy -plane defined by the equation $(x + 10)^2 + (y - 3)^2 = 144$?

- A) $(10, -3)$
 - B) $(10, 3)$
 - C) $(-10, 3)$
 - D) $(-10, -3)$
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Q9. • EASY GRID-IN

A circle in the xy -plane is centered at the origin and passes through the point $(5, 12)$. The equation of the circle can be written as $x^2 + y^2 = c$, where c is a constant. What is the value of c ?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q10. • EASY

A circle in the xy -plane is defined by the equation $(x - 7)^2 + (y - 7)^2 = 100$. What is the area of this circle, in terms of π ?

- A) 10π
 - B) 20π
 - C) 50π
 - D) 100π
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Q11. • MEDIUM

What is the radius of the circle in the xy -plane defined by the equation $x^2 + y^2 - 6x + 8y - 11 = 0$?

- A) 6
 - B) 36
 - C) 11
 - D) 5
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Q12. • MEDIUM

What is the center of the circle in the xy -plane defined by the equation $x^2 + y^2 + 10x - 2y + 13 = 0$?

- A) (5, -1)
 - B) (-5, 1)
 - C) (-5, -1)
 - D) (5, 1)
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Q13. • MEDIUM GRID-IN

What is the radius of the circle in the xy -plane defined by the equation $x^2 + y^2 - 4x - 12y + 4 = 0$?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q14. • MEDIUM

The endpoints of a diameter of a circle in the xy -plane are $(-1, 4)$ and $(7, -2)$. What is the center of the circle?

- A) (6, 2)
 - B) (3, -1)
 - C) (3, 1)
 - D) (-3, 1)
-

Q15. • MEDIUM

The endpoints of a diameter of a circle in the xy -plane are $(-1, 4)$ and $(7, -2)$. What is the radius of the circle?

- A) 10
 - B) 50
 - C) 8
 - D) 5
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Q16. • MEDIUM

The endpoints of a diameter of a circle in the xy -plane are $(-1, 4)$ and $(7, -2)$. Which of the following is an equation of the circle?

- A) $(x - 3)^2 + (y - 1)^2 = 25$
 - B) $(x + 3)^2 + (y + 1)^2 = 25$
 - C) $(x - 3)^2 + (y - 1)^2 = 100$
 - D) $(x - 3)^2 + (y - 1)^2 = 5$
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Q17. • MEDIUM GRID-IN

The equation $x^2 + y^2 + 6x - 4y + c = 0$ defines a circle in the xy -plane with radius 7. What is the value of c ?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q18. • MEDIUM

A circle in the xy -plane is defined by the equation $(x - 4)^2 + (y + 1)^2 = 50$. Point P has coordinates $(10, -1)$. Which of the following describes the location of point P relative to the circle?

- A) P lies on the circle.
 - B) P lies inside the circle.
 - C) P lies outside the circle.
 - D) P's location cannot be determined.
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Q19. • MEDIUM

In the xy -plane, circle A has equation $(x + 6)^2 + (y - 2)^2 = 16$. Circle B has the same center as circle A but has a radius that is 3 times the radius of circle A. Which equation represents circle B?

- A) $(x + 6)^2 + (y - 2)^2 = 48$
 - B) $(x + 6)^2 + (y - 2)^2 = 12$
 - C) $(x + 6)^2 + (y - 2)^2 = 144$
 - D) $(x + 6)^2 + (y - 2)^2 = 19$
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Q20. • MEDIUM

What is the area, in terms of π , of the circle in the xy -plane defined by the equation $x^2 + y^2 - 8x + 2y - 8 = 0$?

- A) 5π
 - B) 10π
 - C) 50π
 - D) 25π
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Q21. • MEDIUM

A sprinkler is installed at the point $(12, 9)$ on a diagram of a backyard, where each unit represents 1 foot. The sprinkler can water any point that is within 15 feet of its location. Which equation represents the boundary of the region the sprinkler can reach?

- A) $(x - 12)^2 + (y - 9)^2 = 225$
 - B) $(x - 12)^2 + (y - 9)^2 = 15$
 - C) $(x + 12)^2 + (y + 9)^2 = 225$
 - D) $(x - 9)^2 + (y - 12)^2 = 225$
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Q22. • MEDIUM GRID-IN

A circle in the xy -plane is defined by the equation $(x - 9)^2 + (y + 5)^2 = 64$. If the center of the circle is (h, k) and the radius is r , what is the value of $h + k + r$?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q23. • MEDIUM

The equation $x^2 + y^2 - 2x - 14y + F = 0$ defines a circle in the xy -plane with radius 9. What is the value of F ?

- A) 31
 - B) -31
 - C) 131
 - D) -131
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Q24. • MEDIUM

What are the x -intercepts of the circle in the xy -plane defined by the equation $(x - 5)^2 + y^2 = 9$?

- A) $(5, 0)$ and $(0, 0)$
 - B) $(-2, 0)$ and $(-8, 0)$
 - C) $(2, 0)$ and $(8, 0)$
 - D) $(3, 0)$ and $(-3, 0)$
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Q25. • MEDIUM

A circle in the xy -plane is defined by the equation $(x - 2)^2 + (y + 3)^2 = 49$. The circle is translated 5 units to the right and 2 units up. Which equation represents the translated circle?

- A) $(x + 7)^2 + (y - 1)^2 = 49$
 - B) $(x - 4)^2 + (y - 2)^2 = 49$
 - C) $(x - 7)^2 + (y + 1)^2 = 7$
 - D) $(x - 7)^2 + (y + 1)^2 = 49$
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Q26. • MEDIUM GRID-IN

What is the diameter of the circle in the xy -plane defined by the equation $x^2 + y^2 + 12x - 16y + 19 = 0$?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q27. • MEDIUM

The graph of $x^2 + y^2 = k$ in the xy -plane is a circle. For which value of k is the radius of the circle equal to $7/2$?

- A) $49/4$
 - B) $7/2$
 - C) $49/2$
 - D) 7
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Q28. • MEDIUM

For how many integer values of r between 1 and 10, inclusive, does the circle $x^2 + y^2 = r^2$ have a circumference greater than 40?

- A) 3
 - B) 4
 - C) 5
 - D) 6
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Q29. • HARD

A circle in the xy -plane has center $(-4, 6)$ and is tangent to the x -axis. What is the radius of the circle?

- A) 4
 - B) 10
 - C) 6
 - D) -4
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Q30. • HARD

In the xy -plane, circle A has equation $(x - 1)^2 + (y - 1)^2 = 9$, and circle B has equation $(x - 9)^2 + (y - 7)^2 = 16$. Which statement best describes the relationship between circle A and circle B?

- A) The circles intersect at exactly two points.
 - B) The circles are externally tangent.
 - C) One circle lies entirely inside the other.
 - D) The circles do not intersect; each lies entirely outside the other.
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Q31. • **HARD**  **GRID-IN**

What is the radius of the circle in the xy -plane defined by the equation $x^2 + y^2 - 22x + 10y + 97 = 0$?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q32. • **HARD**

A circle with radius 13 is tangent to the y -axis and has its center in the second quadrant at the point $(h, 4)$. What is the value of h ?

- A) 13
 - B) -13
 - C) 4
 - D) -4
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Q33. • **HARD**

A circle in the xy -plane has equation $(x + 2)^2 + (y - 5)^2 = r^2$ and is tangent to the line $y = 11$. What is the value of r^2 ?

- A) 6
 - B) 36
 - C) 121
 - D) 25
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Q34. • **HARD**

What is the radius of the circle in the xy -plane defined by the equation $2x^2 + 2y^2 - 8x + 24y + 30 = 0$?

- A) 25
 - B) 10
 - C) 5
 - D) 2.5
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Q35. • **HARD**  **GRID-IN**

The equation $3x^2 + 3y^2 + 30x - 18y + 39 = 0$ defines a circle in the xy -plane. When the equation is written in the form $(x - h)^2 + (y - k)^2 = r^2$, what is the value of r^2 ?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q36. • **HARD**

For what value of k does the equation $x^2 + y^2 - 14x + 6y + k = 0$ represent a circle consisting of exactly one point?

- A) 0
 - B) -58
 - C) 29
 - D) 58
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Q37. • **HARD**

A circular garden has an area of 144π square feet. The garden is centered at the point $(20, 35)$ on a site map, where each unit represents 1 foot. A sprinkler head sits at the point $(20, 23)$ on the same map. Which statement is true about the sprinkler head's position relative to the garden's boundary?

- A) The sprinkler head is on the boundary of the garden.
 - B) The sprinkler head is inside the garden.
 - C) The sprinkler head is outside the garden.
 - D) The sprinkler head's position cannot be determined.
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Q38. • **HARD**

A circle in the xy -plane has equation $(x - 3)^2 + (y + 2)^2 = 100$. A chord of the circle lies along the horizontal line $y = 6$. What is the length of the chord?

- A) 6
 - B) 12
 - C) 18
 - D) 20
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Q39. • **HARD** **GRID-IN**

A circle in the xy -plane has equation $(x + 4)^2 + (y - 1)^2 = 169$. A chord of the circle lies along the vertical line $x = 1$. What is the length of the chord?

GRID-IN — enter your answer as an integer, decimal, or fraction. No units, symbols, or commas.

Q40. • **HARD**

Two concentric circles share the center $(6, -8)$. The larger circle has equation $(x - 6)^2 + (y + 8)^2 = 169$, and the smaller circle has radius 5. What is the area of the region inside the larger circle but outside the smaller circle?

- A) 169π
- B) 64π
- C) 144π
- D) 194π

— END OF QUESTIONS —